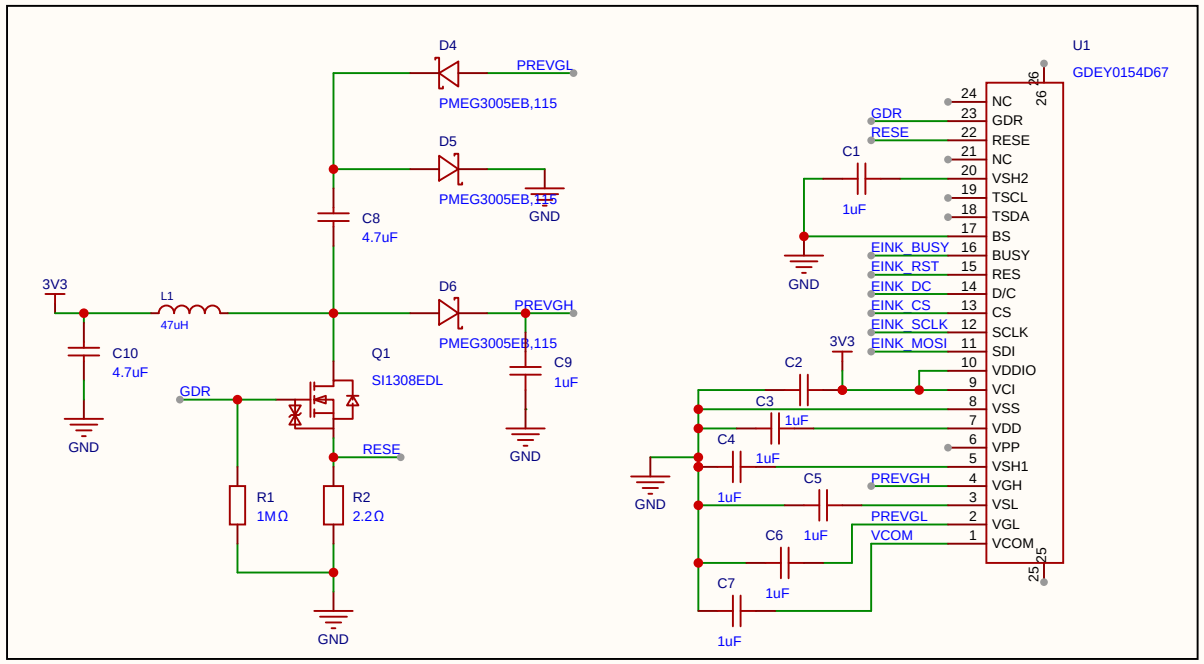
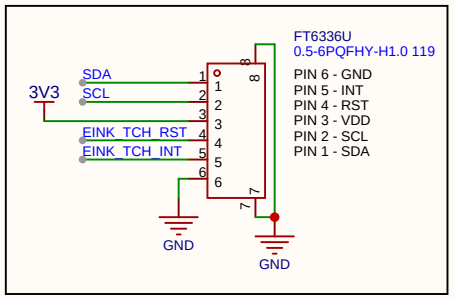
SCREEN

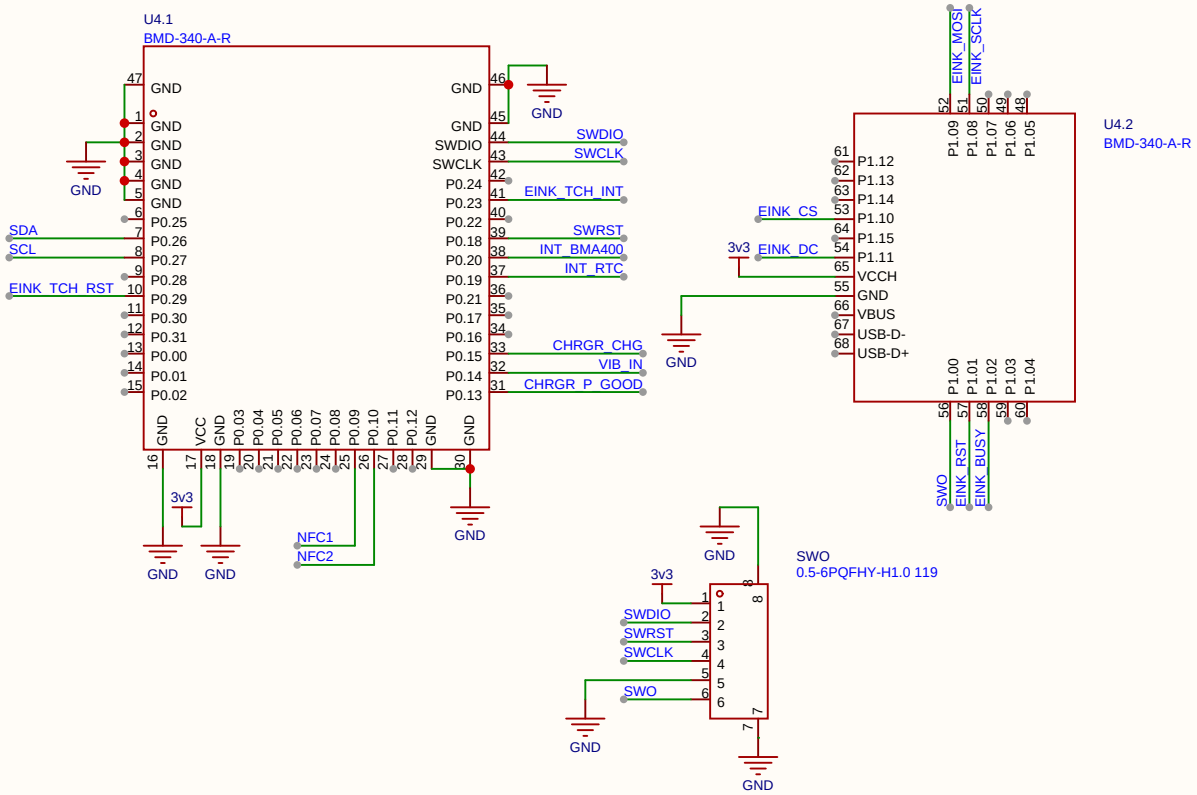


TOUCH



CHECK AGAIN PAGE 33
https://www.good-display.com/public/html/pdfjs/viewer/viewernew.html?file=https://v4.cecdn.yun300.cn/100001_1909185148/GDEY0154D67-T03.pdf

Schematic	Mainboard	Create at	2026-08-30
		Update at	2026-10-23
Board	Mainboard	Page	GDEY0154D67
Drawn		OpenWatch_v1	
Reviewed			
		Version	Size / Page 1 Total 5
EasyEDA		V1.0	A4 / EasyEDA.com



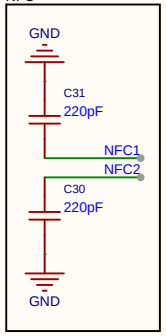
XTAL BLE

XC1 and XC2 are not exposed in BMD-340, it already includes that high-frequency crystal internally

XTAL RTC

Big problem, the proper xtal with same pf doesn't seem to be in stock, and ppm accuracy would be something like 20ppm anyways and battery gain is negligible, going with a RV-3032-C7

NFC

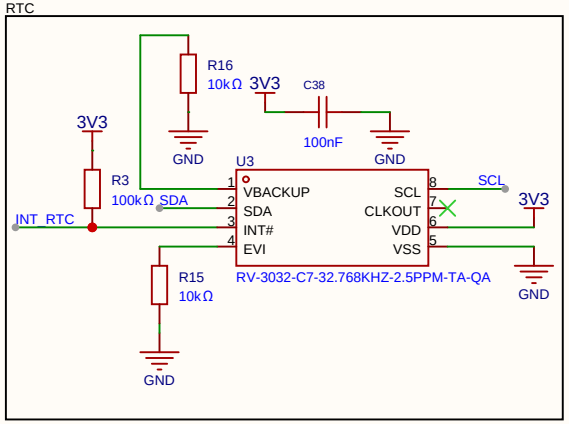


Schematic	Mainboard	Create at	2026-08-30
		Update at	2026-10-23
Board	Mainboard	Page	NRF52840
Drawn		OpenWatch_v1	
Reviewed			
		Version	Size / Page 2 Total 5
EasyEDA		V1.0	A4 / EasyEDA.com

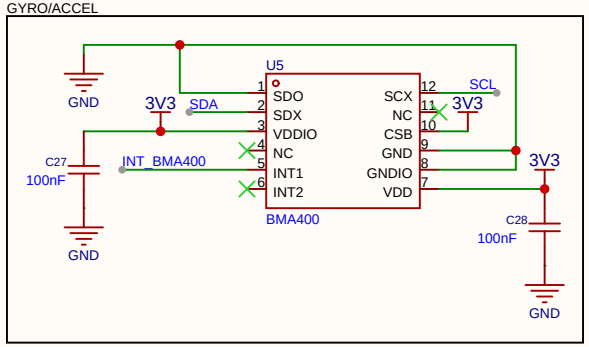
Always powered through VDD pin, with 45nA consumption. Enabling switching to vbackup would increase power consumption.

Disable CLKOUT in software

RTC

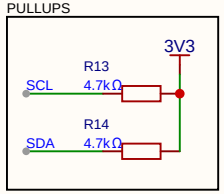


GYRO/ACCEL



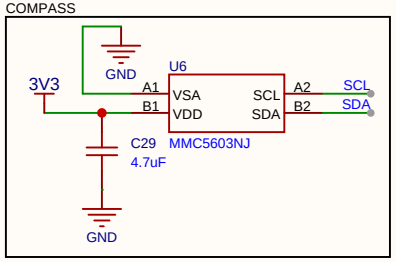
Only interrupt 1 used as only useful feature is wake up on tap

PULLUPS



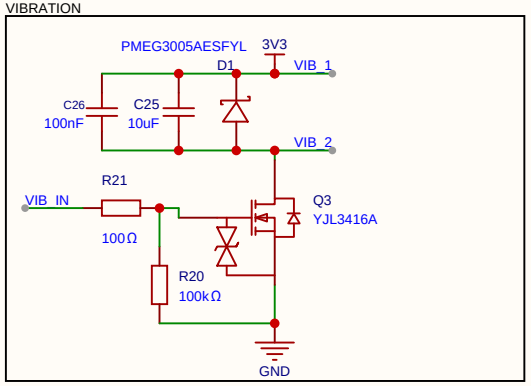
INT PULLUPS NOT NEEDED ON THESE, USING INTERNAL 13KOHM ONES, POWER LOSS NEGLIGIBLE

COMPASS

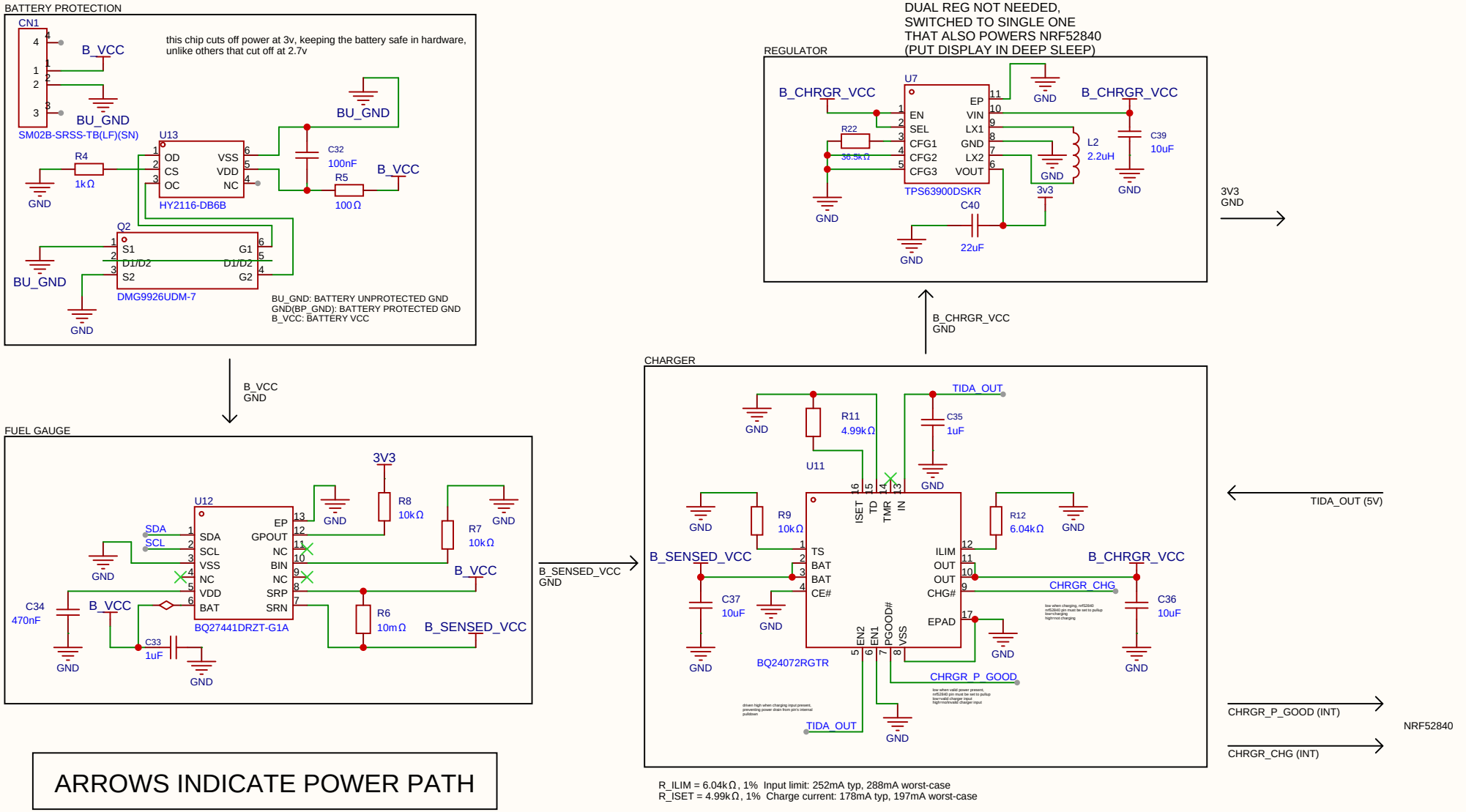


Using 2 caps for high and low frequencies

VIBRATION



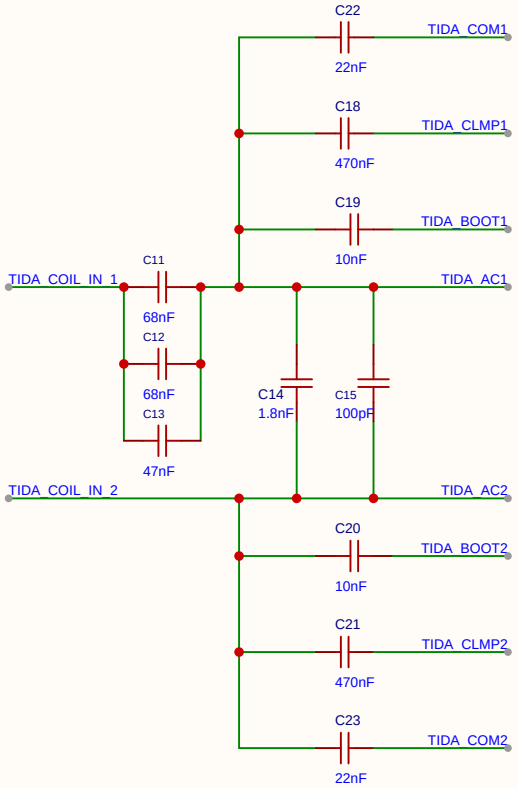
Schematic	Mainboard	Create at	2026-08-29
		Update at	2026-10-23
Board	Mainboard	Page	PERIPHERALS
Drawn		OpenWatch_v1	
Reviewed			
		Version	Size / Page 3 Total 5
EasyEDA		V1.0	A4 / EasyEDA.com

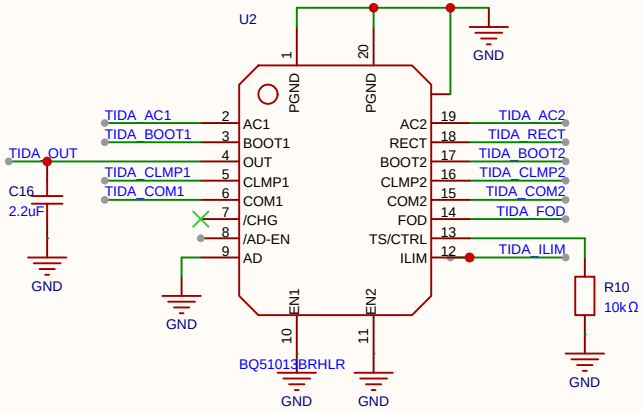


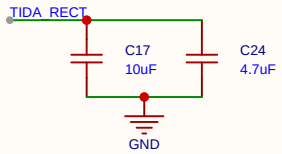
Schematic	Mainboard	Create at	2026-08-30
		Update at	2026-08-30
Board	Mainboard	Page	POWER
Drawn		OpenWatch_v1	
Reviewed			
		Version	Size	Page 4 Total 5
EasyEDA		V1.0	A4	EasyEDA.com

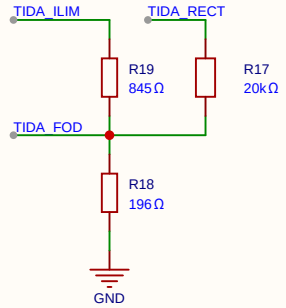
TIDA-00318 qi reference design

Design based on TI TIDA-00318, using different ic (BQ51013B) that is pin compatible for better footprint. link: https://www.ti.com/lit/df/tidraa3/tidraa3.pdf









Schematic	Mainboard	Create at	2026-08-29
		Update at	2026-09-03
Board	Mainboard	Page	TIDA-00318
Drawn		OpenWatch_v1	
Reviewed			
		Version	Size	Page 5 Total 5
EasyEDA		V1.0	A4	EasyEDA.com